

Maize

Zea mays — Pakistan's Fast-Growing Cereal and Feed Crop

A Complete Cultivation Guide for

Punjab · Sindh · Khyber Pakhtunkhwa · Balochistan

Compiled for Farmer Guidance & AI Advisory Use

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1. Introduction

Maize, known locally as makai, has grown rapidly in importance in Pakistan's cropping systems over recent decades, driven mainly by strong demand from the poultry feed industry and increasing adoption of high-yielding hybrid seed. It is grown in both spring and autumn seasons in Punjab and Khyber Pakhtunkhwa, and to a smaller extent in Sindh and Balochistan, making it one of the more flexible cereal crops in the national cropping calendar. Maize is valued for its high yield potential per unit area and its use as food, animal feed, and increasingly, as a raw material for industries such as starch and cooking oil. This guide explains how maize is grown across Pakistan's provinces and how yield can be improved.

2. Scientific Name

Maize's scientific name is *Zea mays*, belonging to the Poaceae (grass) family, the same family as wheat, rice, and sugarcane.

3. Importance in Pakistan

Maize has become Pakistan's third most important cereal crop after wheat and rice, with production expanding rapidly due to the rise of hybrid maize seed and strong demand from the poultry and livestock feed industry. Khyber Pakhtunkhwa has traditionally been a major maize-growing region, especially for local white maize varieties, while Punjab has seen the fastest growth in hybrid yellow maize cultivation linked to commercial poultry feed mills. Maize's relatively short duration and flexibility as both a spring and autumn crop also make it a useful option for farmers seeking to diversify or intensify their cropping system within the year.

4. Major Producing Provinces and Districts

- Punjab: Sahiwal, Okara, Kasur, Chiniot, and Faisalabad — the fastest-growing hybrid maize belt, closely linked to nearby poultry feed industries.
- Khyber Pakhtunkhwa: Swabi, Mardan, Peshawar valley, and Charsadda — Pakistan's traditional maize heartland, especially for local white maize consumed directly as food.
- Sindh: Limited but growing cultivation in some irrigated districts.
- Balochistan: Small-scale cultivation, mainly in areas with reliable irrigation such as Naseerabad division.

5. Climate Requirements

Maize grows best with temperatures between 21°C and 27°C during the vegetative and reproductive stages, and it is sensitive to both very high heat during flowering (which can cause poor pollination and "blank" cobs) and frost, which can damage or kill young plants. Adequate moisture during the critical tasseling and silking stage is essential, since water stress at this time sharply reduces grain set and yield.

6. Soil Requirements

Maize grows best in well-drained, fertile loam to silt loam soils rich in organic matter, with a soil pH between 6.0 and 7.5. It has a relatively high nutrient demand and is less tolerant of poor drainage and waterlogging than rice, so fields prone to standing water should be avoided or improved with proper leveling and drainage channels.

7. Land Preparation

Land is typically ploughed two to three times followed by planking to prepare a fine, well-leveled seedbed, since good early root establishment is important for this fast-growing crop. Farmyard manure, where available, is commonly incorporated ahead of sowing, particularly for hybrid maize crops where farmers are investing in higher input levels to capture the full yield potential of hybrid seed.

8. Recommended Varieties (Pakistan)

- Hybrid maize varieties from private seed companies, widely grown in Punjab for their high yield potential, particularly for the commercial poultry feed market.
- Local white maize varieties and improved open-pollinated varieties developed by the Maize and Millets Research Institute (Yousafwala, Punjab) and the Cereal Crops Research Institute in Khyber Pakhtunkhwa, especially valued in KP for direct food use.
- Farmers should check the latest hybrid and open-pollinated variety recommendations each season from provincial agriculture departments, since hybrid seed must be purchased fresh every year to maintain yield advantages, unlike open-pollinated varieties which can be saved from the farmer's own harvest.

9. Seed Rate

Recommended seed rate for hybrid maize is about 8–10 kg per acre, while local and open-pollinated varieties may require a somewhat higher rate of 10–12 kg per acre due to generally lower seed vigor and germination compared to modern hybrids.

10. Seed Treatment

Seed should be treated with a fungicide such as thiram or carbendazim (2–3 g per kg seed) before sowing to protect against seedling blight and soil-borne diseases; many hybrid seed lots are already pre-treated by the seed company, in which case no additional treatment is generally needed.

11. Sowing Time (Province-wise)

- Punjab: Spring crop sown mid-January to February; autumn crop sown mid-July to August.
- Khyber Pakhtunkhwa: Spring crop sown February–March; autumn crop sown July–August, aligned with the province's traditional two-season maize cycle.
- Sindh: Sown February–March and August, where cultivated.
- Balochistan: Sown March–April in irrigated areas.

12. Sowing Methods

Maize is generally sown by drilling seed in rows about 60–75 cm apart with plant-to-plant spacing of 20–25 cm, at a depth of about 5 cm, using a seed drill or manual dibbling on smaller farms. Proper plant population is important since overcrowding reduces individual cob size and can increase disease pressure, while too sparse a stand wastes the land's yield potential.

13. Fertilizer Recommendations (NPK)

A general recommendation for hybrid maize is 150–180 kg Nitrogen, 90–100 kg Phosphorus (P₂O₅), and 60–75 kg Potash (K₂O) per acre, reflecting the crop's high nutrient demand for its substantial yield potential; local varieties can be managed with somewhat lower rates. Full phosphorus and potash and about one-third of nitrogen are applied at sowing, with the remaining nitrogen split into two or three top-dressings at the knee-high stage, before tasseling, and at silking, since nitrogen availability during these stages strongly influences final grain yield.

14. Irrigation Schedule

Maize requires regular irrigation approximately every 8–12 days depending on soil type and season, with the most critical irrigations at the knee-high stage, tasseling/silking, and grain-filling stages; water stress during tasseling and silking in particular can cause poor pollination and significant yield loss. Waterlogging should be avoided since maize roots are sensitive to prolonged standing water, unlike rice.

15. Weed Management

Maize is a wide-row crop that allows weeds to establish easily between rows if not controlled, so weed management in the first 30–40 days is critical, typically through one to two hoeings or inter-row cultivation. Pre-emergence and post-emergence herbicides approved for maize are increasingly used on larger commercial farms to reduce labor costs, particularly for the fast-expanding hybrid maize sector in Punjab.

16. Insect Pests

Maize stem borer and the invasive fall armyworm (which has spread rapidly across Pakistan's maize-growing areas in recent years) are the most significant insect pests, capable of causing severe damage to both leaves and developing cobs if not managed promptly. Aphids and cutworms can also cause damage at various growth stages. Integrated management includes regular field scouting, timely sowing to avoid peak pest pressure, resistant or tolerant hybrids where available, and need-based insecticide application, with fall armyworm requiring particularly vigilant early detection since it can spread and damage crops quickly.

17. Diseases

Maize leaf blight, common rust, and stalk rot are among the more commonly reported diseases in Pakistan's maize crop, generally worsened by high humidity, dense planting, and excessive nitrogen without balanced potassium. Downy mildew can also be a serious problem in some regions. Management relies on resistant hybrids, balanced fertilization, proper plant spacing for airflow, crop rotation, and fungicide application where disease pressure is confirmed to be high.

18. Deficiency Symptoms

Nitrogen deficiency shows as a distinctive yellowing (chlorosis) starting from the tip of older leaves and moving along the midrib in a V-shaped pattern, along with stunted growth. Phosphorus deficiency causes purplish discoloration on leaves, especially in young plants, and delayed maturity. Potassium deficiency appears as yellowing and browning (firing) along leaf margins, starting with older leaves, along with weak stalks prone to lodging. Zinc deficiency, common in Pakistan's alkaline soils, shows as broad pale stripes along the leaf and is generally corrected with zinc sulfate application.

19. Harvesting

Maize is ready for harvest when the husks turn brown and dry and the kernels harden, with a visible "black layer" forming at the kernel base indicating physiological maturity, typically 90–120 days after sowing depending on variety and season. Harvesting can be done by hand, picking cobs individually, or with combine harvesters on larger mechanized farms, with the crop generally allowed to dry further in the field or in cribs before shelling to reduce grain moisture to safe storage levels.

20. Storage

Shelled maize grain should be dried to a safe moisture level (below about 13–14%) before storage to prevent mold growth (including dangerous aflatoxin-producing molds) and insect infestation. Grain is typically stored in clean, dry, well-ventilated bags or bins, with proper drying being especially critical for maize compared to some other cereals given its susceptibility to mold-related quality and safety problems if stored at high moisture.

21. Expected Yield in Pakistan

The national average maize yield in Pakistan has risen substantially with the spread of hybrid seed, now averaging around 4.5–5 tons per hectare, though this masks a wide range between hybrid maize grown under good management in Punjab (which can exceed 8–10 tons per hectare) and traditional local varieties grown with fewer inputs in parts of Khyber Pakhtunkhwa, which yield considerably less.

22. Government Recommendations

The Maize and Millets Research Institute at Yousafwala and the Cereal Crops Research Institute in Khyber Pakhtunkhwa lead variety development and agronomic research for maize, working with provincial extension departments to promote certified seed use, balanced fertilization, and integrated pest management, particularly for the fall armyworm threat. Government policy has also supported the growing

maize-poultry feed value chain given its importance to the broader livestock and food security sectors.

23. Modern Farming Practices

The rapid shift toward hybrid maize seed, combined with balanced fertilizer use and improved plant population management, represents the most significant modern practice change in Pakistan's maize sector over the past two decades. Mechanized planting and harvesting are increasingly common on larger commercial farms in Punjab, alongside growing use of certified seed and recommended fertilizer packages tailored to hybrid maize's higher input responsiveness.

24. Precision Agriculture

Precision practices being introduced in maize include soil testing to guide nitrogen, phosphorus, and zinc application, and variable rate fertilizer application on some larger farms to match inputs to field variability. Mobile-based pest alert systems, particularly important given the fall armyworm threat, and satellite-based crop monitoring are being extended to maize-growing areas by research institutes and some private sector partners.

25. Sustainable Farming

Sustainable maize practices being promoted in Pakistan include balanced fertilization guided by soil testing to avoid excess nitrogen use (which increases both cost and disease risk), integrated pest management to reduce reliance on insecticides against fall armyworm and other pests, and crop rotation with legumes or other crops to maintain soil health and break pest and disease cycles associated with continuous maize cultivation.

26. Regional Notes

Khyber Pakhtunkhwa remains the traditional heartland of Pakistani maize, particularly for local white maize used directly as food, while Punjab has emerged as the fastest-growing hybrid maize region driven by strong linkages to the commercial poultry feed industry. As fall armyworm and other pest pressures continue to challenge the crop, timely adoption of resistant hybrids, proper scouting, and integrated pest management will remain central to sustaining maize's rapid growth as one of Pakistan's most dynamic cereal crops.